

Block Influence in Deep Residual Networks: Silent Representational Degradation, Block Interaction, and a Null Result for BI-Weighted Knowledge Distillation

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Abstract—Depth-pruning criteria in the ShortGPT family score a layer by a cheap redundancy proxy, typically input-output similarity, and validate it against end-task accuracy, implicitly assuming that behavioral invariance implies representational invariance. This paper measures where that assumption fails and whether it matters for distillation. A ResNet-50 teacher on CIFAR-10 (test accuracy 0.9464) provides 16 residual blocks, each scored by three Block Influence (BI) metrics: passive geometric transformation (BI_{geo}), accuracy drop under ablation (BI_{acc}), and representational disruption under ablation via linear centered kernel alignment (BI_{rep}). The metrics agree on block ranking (Kendall’s $\tau = 0.63$ – 0.85) but dissociate locally: ablating layer4.0 reduces accuracy by only 0.037 yet causes the network’s largest representational damage. On samples still classified correctly, softmax entropy rises 19-fold and class centroids merge. Real multi-block pruning shows joint damage is about twice the additive prediction at $k = 9$. Weighting similarity-preserving distillation by per-stage BI left a ResNet-18 student statistically unchanged; the null result traces to stage aggregation and loss-share imbalance, not the weighting hypothesis. Geometric fidelity to the teacher conferred no discriminative advantage: the least faithful student achieved the highest class separation and the best corruption robustness.

Index Terms—Depth pruning, block ablation, centered kernel alignment, representational similarity, knowledge distillation, residual networks, silent failure.

I. INTRODUCTION

DEPH pruning removes entire blocks from trained residual networks to obtain smaller models without retraining. Criteria in the ShortGPT family [1] score a layer by a cheap redundancy proxy, typically the cosine similarity between a block’s input and output representations, rank blocks by that score, and validate the resulting removal schedule against end-task accuracy. The implicit assumption is that behavioral invariance implies representational invariance: if accuracy remains unchanged, nothing important was lost.

This assumption fails in a specific and predictable way for classification heads over high-dimensional feature spaces. Accuracy is a one-bit-per-sample readout of a 2048-dimensional

representation. A block can be removed, the argmax can survive on $\sim 96\%$ of samples, and yet the geometry the classifier reads from (margins, class-centroid structure, within-class dispersion) can be substantially degraded. Any downstream consumer of the representation other than the original argmax (transfer, calibration, out-of-distribution behavior, a distillation student) inherits damage that the accuracy metric never registered. We call a block with this profile a *silent-failure* candidate: low accuracy damage under ablation, high representational damage.

This project asks two questions. *Q1 (measurement)*: do accuracy-based and representation-based importance measures agree at the block level? Where they dissociate, is the dissociation behaviorally real or an artifact of the representational measure? *Q2 (prescription)*: does the difference matter in practice? Specifically, does weighting a similarity-preserving distillation loss [2] by per-block influence measures produce a better student than uniform weighting?

To answer Q1, each of the 16 Bottleneck blocks of a ResNet-50 trained on CIFAR-10 is scored by three metrics: BI_{geo} (how much the block transforms its input, measured passively), BI_{acc} (accuracy drop under ablation), and BI_{rep} (representational disruption under ablation, measured by linear CKA on pre-classifier features). A tercile-based criterion flags blocks where BI_{rep} is high while BI_{acc} is low, ranked by the discrepancy $\Delta = BI_{rep} - BI_{acc}$; every flagged block then undergoes a behavioral confirmation that involves no CKA in the measurement loop. To answer Q2, a ResNet-18 student is trained under four conditions (vanilla, uniform SP-KD, BI_{acc} -weighted, BI_{rep} -weighted; three seeds each) and evaluated on accuracy, representational fidelity, fine-grained geometry, confidence, calibration, and robustness to distribution shift.

The contributions are as follows.

- 1) A measurement pipeline computing three Block Influence metrics per block, with a silent-failure criterion (Δ) designed to be robust to the known failure mode of CKA [4] through triangulation and behavioral confirmation.
- 2) Identification and behavioral confirmation of a silent-failure block (layer4.0): ablation reduces accuracy by 0.037 while softmax entropy on still-correct samples rises 19-fold and class-centroid pairs merge.
- 3) Evidence from real multi-block pruning that marginal (single-block) scores understate joint damage by $\sim 2.2\times$

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at $k = 9$ removed blocks, a superadditivity limitation shared with all marginal criteria, including ShortGPT-style scores.

- 4) A controlled null result for BI-weighted SP-KD with a mechanistic diagnosis: stage-averaging collapses the two weight vectors to near-identity (Pearson $r = 0.999$) and the weighted loss term is $\sim 99\%$ of the objective, so the mechanism under test had no gradient leverage.
- 5) Evidence that teacher geometry is not a normative target for a smaller student: the least faithful student roughly doubles the teacher’s class separation (FDR 15.7 vs. 7.6) and is the most corruption-robust.

Section II reviews related work; Section III defines the metrics, the silent-failure criterion, and the distillation design; Section IV reports results; Sections V and VI discuss implications and the follow-up experiment.

II. BACKGROUND AND RELATED WORK

A. Depth pruning and marginal importance criteria

ShortGPT [1] popularized block-level redundancy scoring for large language models: a Block-Influence score based on the similarity between a block’s input and output, validated by removing the lowest-scoring layers and measuring end-task performance. The validation target is always behavioral, and the score is marginal: each block is assessed in isolation. Section IV-C shows that marginality itself is a limitation (joint damage grows superadditively), and Section IV-A shows that behavioral validation can miss real damage entirely.

B. Representational similarity and its caveats

Linear CKA [3] compares two representations through the similarity of their Gram matrices and is the standard tool for measuring representational change. Ding *et al.* [4] showed that CKA is dominated by the high-variance principal components: large CKA changes can be driven by task-irrelevant directions, and changes confined to low-variance but class-informative directions can be missed. Our criterion therefore never interprets CKA magnitude as importance in isolation; it uses CKA only as one term of a disagreement signal, triangulated with measures of different failure modes and confirmed behaviorally (Section III-C).

The measurement framing is the model-side analogue of representational similarity analysis in systems neuroscience [5], and of the goal-driven modeling program of Yamins and DiCarlo [6]. Block ablation is a virtual lesion; a silent-failure block is the model analogue of lesion dissociations in which overt task behavior survives while the underlying representational code degrades, as in associative agnosia [7], where recognition may succeed while the internal structure of the percept is measurably impoverished.

C. Knowledge distillation

Hinton *et al.* [8] introduced logit-matching distillation; Tung and Mori [2] proposed similarity-preserving KD (SP-KD), which matches batchwise activation Gram matrices between teacher and student layers; Tian *et al.* [9] framed distillation

as contrastive representation learning. Higher teacher fidelity does not automatically translate into better students [10]; our Phase-4 results provide a clear instance of that dissociation, plus a participation-ratio diagnosis of what full-Gram matching does to a student’s effective dimensionality.

III. METHODS

A. Models, data, and features

The measurement target is a ResNet-50 [11] with a CIFAR stem (3×3 stride-1 convolution, no max-pooling), 23.52M parameters, and test accuracy 0.9464 on CIFAR-10 [12] (weights: edadaltocg/resnet50_cifar10; verified against the published 0.9465). Let f denote the intact network and $f_{-\ell}$ the network with block ℓ ablated, implemented as an identity forward pass; for the four transition blocks (layer{1,2,3,4}.0) the downsample projection branch is preserved so that tensor shapes remain valid. All representation metrics are computed on a calibration set of $N = 2500$ training images (250 per class, rebuilt deterministically) and GAP-pooled pre-classifier features $F \in \mathbb{R}^{N \times 2048}$; BI_{acc} is computed on the full 10,000-image test set.

B. Block Influence metrics

Column-centered linear CKA [3] between feature matrices $X, Y \in \mathbb{R}^{N \times D}$ is

$$\text{CKA}(X, Y) = \frac{\|Y^\top X\|_F^2}{\|X^\top X\|_F \|Y^\top Y\|_F}. \quad (1)$$

Each block receives three primary scores and two auxiliary ones.

1) BI_{geo} : passive geometric transformation:

$$\text{BI}_{\text{geo}}(\ell) = 1 - \frac{1}{N} \sum_{i=1}^N \cos(\text{GAP}(x_{\ell, \text{in}}^{(i)}), \text{GAP}(x_{\ell, \text{out}}^{(i)})), \quad (2)$$

where $x_{\ell, \text{in}}, x_{\ell, \text{out}}$ are the block’s input and output maps (for transition blocks the input is passed through the block’s own downsample projection so both sides share dimensionality). $\text{BI}_{\text{geo}} \approx 0$ means the block barely transforms its input, redundant in the ShortGPT sense, and the metric costs a single forward pass for all blocks.

2) BI_{acc} : behavioral importance:

$$\text{BI}_{\text{acc}}(\ell) = \text{Acc}(f) - \text{Acc}(f_{-\ell}). \quad (3)$$

3) BI_{rep} : representational importance:

$$\text{BI}_{\text{rep}}(\ell) = 1 - \text{CKA}(F_{\text{intact}}, F_{-\ell}). \quad (4)$$

4) *Auxiliary scores*: $\text{BI}_{\text{rep-gram}}$ recomputes (4) through the double-centered Gram formulation, $K = F_c F_c^\top \in \mathbb{R}^{N \times N}$,

$$\text{BI}_{\text{rep-gram}}(\ell) = 1 - \frac{\text{tr}(K_{\text{intact}}^c K_{-\ell}^c)}{\sqrt{\text{tr}(K_{\text{intact}}^c K_{\text{intact}}^c) \text{tr}(K_{-\ell}^c K_{-\ell}^c)}}, \quad (5)$$

which is mathematically equivalent for the linear kernel; it is retained as an implementation cross-check ($|\text{BI}_{\text{rep}} - \text{BI}_{\text{rep-gram}}| < 3 \times 10^{-5}$ on all 16 blocks) and as the extension point for non-linear kernels. $\text{BI}_{\text{rep-cls}}$ targets inter-class angular

Algorithm 1 Silent-failure candidate identification

Require: per-block scores $\{\text{BI}_{\text{acc}}(\ell), \text{BI}_{\text{rep}}(\ell)\}_{\ell=1}^{16}$

- 1: $\theta_{\text{acc}} \leftarrow P_{33}(\{\text{BI}_{\text{acc}}(\ell)\})$; $\theta_{\text{rep}} \leftarrow P_{67}(\{\text{BI}_{\text{rep}}(\ell)\})$
- 2: $C \leftarrow \{\ell : \text{BI}_{\text{acc}}(\ell) < \theta_{\text{acc}} \wedge \text{BI}_{\text{rep}}(\ell) > \theta_{\text{rep}}\}$
- 3: **if** $C = \emptyset$ **then**
- 4: $C \leftarrow \{\arg \max_{\ell} \Delta(\ell)\}$ ▷ null-result fallback
- 5: **end if**
- 6: **return** C ranked by $\Delta(\ell) = \text{BI}_{\text{rep}}(\ell) - \text{BI}_{\text{acc}}(\ell)$

structure: with class centroids $\mu_c = \text{mean}\{F_i : y_i = c\}$ and $S_{ab} = \cos(\mu_a, \mu_b)$ the 10×10 centroid-similarity matrix,

$$\text{BI}_{\text{rep-cls}}(\ell) = 1 - \cos(\text{vec}(S_{\text{intact}}), \text{vec}(S_{-\ell})). \quad (6)$$

Finally, the silent-failure discrepancy is

$$\Delta(\ell) = \text{BI}_{\text{rep}}(\ell) - \text{BI}_{\text{acc}}(\ell). \quad (7)$$

Estimator stability. With $D = 2048$ and $N = 2500$ the calibration set only marginally exceeds the feature dimensionality; $N \approx 3000$ – 5000 would give full-rank stability of the Gram spectrum. $N = 2500$ was chosen to bound compute cost and is an acknowledged limitation.

C. The silent-failure criterion and its triangulation

Block ℓ is a silent-failure candidate if $\text{BI}_{\text{acc}}(\ell)$ falls in the lower tercile of the empirical BI_{acc} distribution and $\text{BI}_{\text{rep}}(\ell)$ in the upper tercile of the BI_{rep} distribution; candidates are ranked by Δ (Algorithm 1). Tercile thresholds adapt to the empirical distribution and guarantee a non-trivial candidate population; if no block satisfies both criteria, the protocol falls back to the highest- Δ block.

Because Ding *et al.* [4] showed that CKA can change for task-irrelevant reasons, a large BI_{rep} alone is not treated as evidence. Three design choices mitigate (not eliminate) the failure mode. (i) Δ is a *disagreement* signal: a block is flagged only where a variance-weighted geometric measure and a task measure dissociate, never CKA magnitude alone. (ii) BI_{rep} is triangulated with two measures of different failure modes, the Gram-route computation (5) and the centroid-level angular measure (6). (iii) Every candidate undergoes behavioral confirmation independent of CKA, namely softmax entropy and top-1 confidence on still-correct samples, tested with one-sided Wilcoxon signed-rank tests. A low-variance, class-critical direction that neither CKA nor the entropy of still-correct predictions detects would still be missed; this is a mitigation strategy, not a refutation of the critique.

D. Pipeline and implementation

The pipeline has four phases with disk-checkpointed hand-offs (Fig. 1); each phase can be re-run in isolation. Seed 42 fixes all random number generators and enables deterministic algorithms for the measurement phases (relaxed to cuDNN benchmark mode for Phase-4 training, where determinism caused a 5–10 $\times$ slowdown; seed-level reproducibility is retained). Ablation is a context manager that patches the block’s forward method and restores it in a `finally` clause; multi-block pruning enters k such contexts through an exit stack

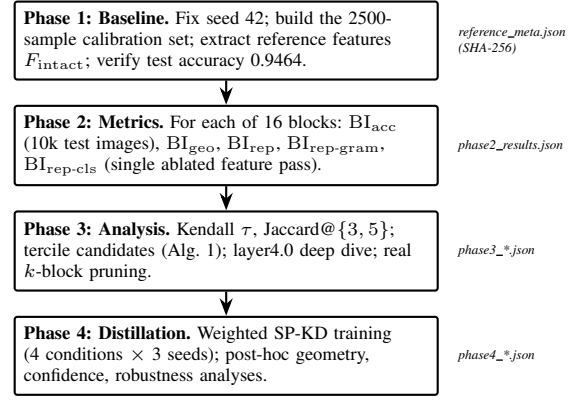


Fig. 1. Pipeline overview. Each phase writes JSON artifacts that are the sole source of the numbers in this paper (Appendix A).

before any forward pass, guaranteeing atomic multi-block ablation and safe restoration. The reference feature tensor is SHA-256 checksummed into its metadata file. Validation relies on runtime checks with concrete thresholds: the intact model’s accuracy must exceed 0.90 (a warning is raised below 0.93), every BI metric must lie in its expected range, a monotonicity spot-check flags implausible BI_{acc} orderings for manual inspection, and the rank-correlation and Jaccard summaries must be mutually consistent. Identity tests verify the robustness metrics (self-mCE = 1.0, perfect-calibration ECE = 0), and the $\text{BI}_{\text{rep}} \approx \text{BI}_{\text{rep-gram}}$ cross-check is retained as an implementation guard. There is no automated test suite in the repository.

E. Phase 4: SP-KD with BI-derived stage weights

1) *Loss:* For a matched layer pair with a batch of N activations, the normalized Gram matrix is $G = \tilde{F}\tilde{F}^\top$ with $\tilde{F}$ the row-normalized flattened activations, and the similarity-preserving loss [2] with stage weights w_s is

$$\mathcal{L}_{\text{SP}} = \sum_{s \in \text{stages}} w_s \frac{\|G_T^{(s)} - G_S^{(s)}\|_F^2}{N^2}, \quad \mathcal{L} = \mathcal{L}_{\text{CE}} + \gamma \mathcal{L}_{\text{SP}}. \quad (8)$$

Because G is $N \times N$, teacher (channels 256/512/1024/2048) and student (64/128/256/512) stages can be matched without projection heads. Matching pairs the last block of each stage: teacher layer1.2/2.3/3.5/4.2 with student layer1.1/2.1/3.1/4.1, verified at runtime by a spatial-dimension check.

2) *The mechanism under test:* ResNet-50’s 16 per-block scores are averaged into 4 stage means, floored at 0.05, and normalized to sum to one (Algorithm 2). This stage aggregation is where the block-level signal is lost (Section IV-D).

3) *Conditions:* Seven conditions were implemented (Table I); the four primary ones were trained to completion with full artifacts (three seeds each). The three Δ -weighted ablations decompose confidence gating and soft-label KD; their artifacts were not archived (Table I, status column), and their outcomes are excluded from all quantitative claims.

4) *Training and evaluation:* The student is a ResNet-18 with the same stem modification (11.17M parameters), trained from scratch with standard augmentation (random crop 32/4,

Algorithm 2 BI-weighted SP-KD stage weighting and loss

Require: block scores $\{\text{BI}(\ell)\}$, stages $\mathcal{S} = \{1, \dots, 4\}$, floor $\epsilon = 0.05$, weight γ

- 1: **for** $s \in \mathcal{S}$ **do**
- 2: $w_s \leftarrow \max(\text{mean}\{\text{BI}(\ell) : \ell \in s\}, \epsilon)$
- 3: **end for**
- 4: $w_s \leftarrow w_s / \sum_t w_t$ $\triangleright$ weights sum to 1
- 5: **return** weights w_s used in the loss of Eq. (8)

TABLE I

PHASE-4 CONDITIONS. “KD-SOFT” IS HINTON DISTILLATION [8] AT $T=4$, $\beta=1$; THE GATE WEIGHTS EACH GRAM ENTRY BY THE TEACHER’S TRUE-CLASS PROBABILITY.

Condition	Loss	Stage weights	Artifacts
vanilla	$\mathcal{L}_{\text{CE}}$	n/a	3 seeds
uniform	$\mathcal{L}_{\text{CE}} + \gamma \mathcal{L}_{\text{SP}}$	1/4 each	3 seeds
bi_acc	$\mathcal{L}_{\text{CE}} + \gamma \mathcal{L}_{\text{SP}}$	$\propto$ stage $\overline{\text{BI}}_{\text{acc}}$	3 seeds
bi_rep	$\mathcal{L}_{\text{CE}} + \gamma \mathcal{L}_{\text{SP}}$	$\propto$ stage $\overline{\text{BI}}_{\text{rep}}$	3 seeds
conf_gated	$\mathcal{L}_{\text{CE}} + \beta \text{KD} + \gamma \mathcal{L}_{\text{SP},g}$	$\propto$ stage $\overline{\Delta}$	records only
gated_only	$\mathcal{L}_{\text{CE}} + \gamma \mathcal{L}_{\text{SP},g}$	$\propto$ stage $\overline{\Delta}$	none
delta_kd	$\mathcal{L}_{\text{CE}} + \beta \text{KD} + \gamma \mathcal{L}_{\text{SP}}$	$\propto$ stage $\overline{\Delta}$	none

horizontal flip), SGD with Nesterov momentum (learning rate 0.1, momentum 0.9, weight decay 5×10^{-4} , batch 128, 200 epochs, step decay $\times 0.1$ at epochs $\{60, 120, 160\}$), $\gamma = 3000$, gradient clipping (norm 5.0) for the first 10 epochs, seeds $\{42, 123, 456\}$. Teacher–student CKA is computed on the calibration set every 20 epochs; a pre-training dry run measures the $\gamma \mathcal{L}_{\text{SP}} / \mathcal{L}_{\text{CE}}$ ratio. Two geometric quantities are used throughout the evaluation:

$$\text{FDR}(a, b) = \frac{\|\mu_a - \mu_b\|^2}{\sigma_a^2 + \sigma_b^2}, \quad \text{PR} = \frac{(\sum_i s_i^2)^2}{\sum_i s_i^4}, \quad (9)$$

where μ_a, μ_b and σ_a^2, σ_b^2 are class-centroid means and variances (FDR over all 45 class pairs and the critical pairs, carried over from the Phase-3 class-pair analysis), and $\{s_i\}$ is the feature covariance spectrum (participation ratio). The remaining evaluation covers test accuracy; per-class CKA and class-cosine structure; intra-class variance; confidence, entropy, logit margin, and expected calibration error (ECE) [14]; CIFAR-10-C (19 corruptions $\times$ 5 severities; mCE with per-corruption normalization by the teacher, so teacher mCE $\equiv 1$; relative mCE) and CIFAR-10-P (10 perturbation sequences; mean flip probability) [13].

IV. RESULTS

A. Metric agreement and its limits

Table II reports all 16 blocks. Three observations. First, the four transition blocks (layer*.0, the only ones that change dimensionality) dominate every metric: importance is concentrated where the representation is re-encoded. Second, the metrics agree globally: Kendall’s $\tau(\text{BI}_{\text{geo}}, \text{BI}_{\text{acc}}) = 0.85$ ($p < 10^{-6}$), $\tau(\text{BI}_{\text{geo}}, \text{BI}_{\text{rep}}) = 0.633$ ($p = 3.4 \times 10^{-4}$), $\tau(\text{BI}_{\text{acc}}, \text{BI}_{\text{rep}}) = 0.683$ ($p = 8.3 \times 10^{-5}$); top-3 and top-5 Jaccard are both 1.0 (top-3: $\{\text{layer1.0}, \text{layer2.0}, \text{layer3.0}\}$; top-5 adds $\{\text{layer1.1}, \text{layer4.0}\}$). Third, the dissociation is real but

TABLE II

PER-BLOCK BLOCK INFLUENCE METRICS (16 BOTTLENECK BLOCKS OF THE RESNET-50 TEACHER). $\Delta = \text{BI}_{\text{rep}} - \text{BI}_{\text{acc}}$. THE SILENT-FAILURE BLOCK LAYER4.0 IS BOLD.

Block	BI_{geo}	BI_{acc}	BI_{rep}	$\text{BI}_{\text{rep-cls}}$	Δ
layer1.0	0.591	0.279	0.366	0.0297	0.087
layer1.1	0.110	0.096	0.137	0.0096	0.041
layer1.2	0.053	0.034	0.064	0.0024	0.030
layer2.0	0.483	0.172	0.228	0.0148	0.056
layer2.1	0.042	0.013	0.018	0.0003	0.005
layer2.2	0.028	0.011	0.017	0.0004	0.006
layer2.3	0.028	0.006	0.013	0.0003	0.007
layer3.0	0.609	0.262	0.313	0.0321	0.052
layer3.1	0.034	0.010	0.018	0.0006	0.008
layer3.2	0.028	0.010	0.017	0.0006	0.007
layer3.3	0.033	0.014	0.023	0.0011	0.009
layer3.4	0.036	0.019	0.034	0.0023	0.015
layer3.5	0.048	0.018	0.043	0.0033	0.024
layer4.0	0.058	0.037	0.135	0.0130	0.098
layer4.1	0.019	0.005	0.040	0.0033	0.035
layer4.2	0.019	0.006	0.047	0.0090	0.041

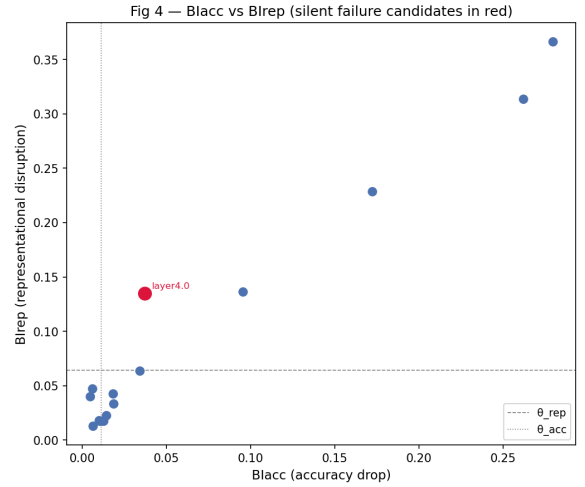


Fig. 2. Silent-failure plane: BI_{acc} (abscissa) vs. BI_{rep} (ordinate); each point is a block; dashed lines are the tercile thresholds θ_{acc} and θ_{rep} . The diagonal cluster shows global metric agreement; layer4.0 (red) is the only block in the low- BI_{acc} /high- BI_{rep} quadrant.

local: layer4.0 is the only block satisfying both tercile criteria ($\text{BI}_{\text{acc}} = 0.037$ below θ_{acc} , $\text{BI}_{\text{rep}} = 0.135$ above θ_{rep}), with the network-largest $\Delta = 0.098$ (Fig. 2). The passive score BI_{geo} gives layer4.0 no distinctive rank (0.058, 11th of 16 in ascending order, within the range of the surrounding redundant blocks): the dissociation is visible only to the ablation-based BI_{rep} .

B. The silent failure of layer4.0

Ablating layer4.0 moves test accuracy from 0.9464 to 0.9091. On the 8974 test samples that remain correctly classified, mean softmax entropy rises from 0.0186 to 0.3543 nats (mean $\Delta H = +0.336$, median $+0.156$; one-sided Wilcoxon $p < 10^{-6}$) and mean top-1 confidence falls from 0.9941 to 0.9057 ($p < 10^{-6}$). The model remains correct on these samples while becoming $\sim 19\times$ more uncertain (Fig. 3). This is the behavioral signature of silent degradation, measured

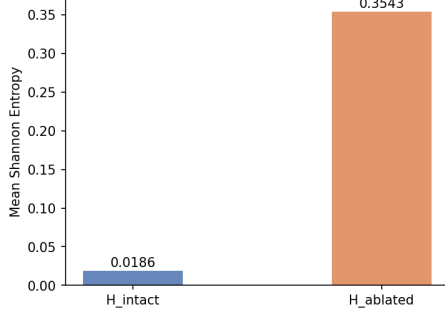
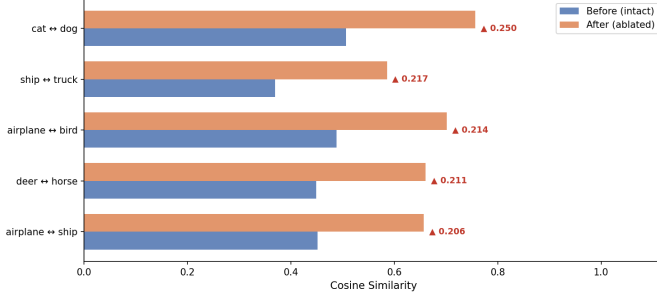
Fig 7 — Entropy Comparison on C₁ (correctly classified samples): layer4.0
 $\Delta H_{\text{mean}}=0.3358$ Wilcoxon $p=0.0000^*$ Fig. 3. Distribution of per-sample softmax entropy on still-correct samples, intact vs. layer4.0-ablated. The mass shift (mean 0.019 $\rightarrow$ 0.354 nats) is the output-level signature of the silent failure: accuracy is preserved while uncertainty increases sharply.Fig 11 — Top-5 Class-Pair Similarity Changes: layer4.0 ablated
All pairs become MORE similar (▲) — class separation degrades silently

Fig. 4. Top-5 class-pair cosine-similarity changes after layer4.0 ablation (blue: intact; orange: ablated). Every top pair becomes more similar after ablation: the block maintains inter-class separation.

entirely at the output level with no CKA in the loop. It rules out the concern that BI_{rep} 's signal for this block is a representational-measure artifact.

The 10×10 class-cosine analysis shows the damage is structured as pair mergers rather than uniform shrinkage. After ablating layer4.0, the centroid cosine similarity rises by +0.250 for cat↔dog ($\approx 0.51 \rightarrow 0.76$), +0.217 for ship↔truck, +0.214 for airplane↔bird, +0.211 for deer↔horse, and +0.206 for airplane↔ship (Fig. 4); all top-5 changes are losses of class separation. Per-class CKA spans 0.472 (bird) to 0.582 (ship) with variance 0.0017: the disruption is broad and relatively uniform across classes. These critical pairs (cat/dog/deer) are carried into Phase 4.

C. Progressive pruning: marginal scores understate joint damage

Single-block scores implicitly assume blocks contribute independently. This prediction fails in the informative direction. For $k = 1, \dots, 16$ blocks ablated simultaneously, in three removal orderings (BI_{acc} -, BI_{rep} -, BI_{geo} -ascending), real representational damage is systematically superadditive in the mid-range. At $k = 9$, real BI_{rep} is 0.523 vs. an additive prediction of 0.237 for the BI_{acc} -ascending order ($2.20\times$), 0.525 vs. 0.224 ($2.35\times$) for BI_{rep} -ascending, and 0.529 vs. 0.228 ($2.32\times$) for BI_{geo} -ascending (Fig. 5). (At $k \geq 14$ the additive sum exceeds 1 while real BI_{rep} saturates near 0.82,

TABLE III

PHASE-4 FINAL METRICS PER CONDITION (SEEDS 42/123/456; MEANS OVER SEEDS). WELCH p IS THE PAIRWISE TEST VS. UNIFORM.

Condition	Accuracy (seed)			Mean	Mean CKA
	42	123	456	acc	(student, teacher)
vanilla	0.9500	0.9488	0.9493	0.9494	0.9387
uniform	0.9519	0.9531	0.9495	0.9515	0.9760
bi_acc	0.9532	0.9528	0.9504	0.9521	0.9791
bi_rep	0.9527	0.9541	0.9494	0.9521	0.9782

so the ratio is only meaningful before saturation.) Accuracy is non-monotone in k (0.428 at $k = 10$ vs. 0.450 at $k = 11$), directly exhibiting block interaction; from $k = 10$ the three orderings converge on the same residual block set and the curves coincide.

Two secondary findings. (a) The entropy-on-correct-samples trace H_{C1} spikes to ~ 1.73 – 1.79 nats around $k = 12$ – 13 , immediately after layer4.0 and layer1.1 are removed, while some samples are still classified correctly. This is a population-level version of the single-block silent failure: the model passes through a regime in which predictions remain correct with almost no margin before collapse. (b) BI_{geo} -ordered and BI_{acc} -ordered pruning produce nearly identical curves: as a pruning guide, the passive score adds little beyond the behavioral one in this setting.

D. Phase 4: the null result and its two mechanical causes

Final metrics are given in Table III. All three SP-KD conditions are statistically indistinguishable in accuracy: pairwise Welch t -tests give $p = 0.97$ (bi_{acc} vs. bi_{rep}), $p = 0.67$ (bi_{acc} vs. uniform), $p = 0.76$ (bi_{rep} vs. uniform), and even SP-KD vs. vanilla does not reach significance with $n = 3$ seeds ($p = 0.07$ – 0.19). SP-KD faithfully optimizes its objective: it copies the teacher's geometry (final CKA 0.976–0.979 vs. 0.939 for vanilla), and none of the effects hypothesized to follow from differential weighting occurred. The null result was largely determined by the experimental design, through two mechanisms.

1) *Cause 1: stage averaging collapses the weighting signal:* The block-level metrics disagree meaningfully ($\tau(\text{BI}_{\text{acc}}, \text{BI}_{\text{rep}}) = 0.683$; $\Delta(\text{layer4.0}) = 0.098$), but after stage-averaging, flooring, and normalization the two weight vectors are

$$\text{bi}_{\text{acc}}: [0.4661, 0.1731, 0.1900, 0.1709]$$

$$\text{bi}_{\text{rep}}: [0.4642, 0.1700, 0.1837, 0.1821]$$

with maximum difference 1.12 percentage points (on layer4), cosine similarity 0.9997, and Pearson $r = 0.9986$. Averaging six blocks of layer3 (one informative, five near-redundant) into one number dilutes precisely the transition-block signal the metrics were built to detect; layer4.0's Δ , the project's central Phase-1–3 finding, contributes one third of one stage mean. The conditions were near-duplicates before training began.

2) *Cause 2: loss-share imbalance makes any normalized weighting inert:* The pre-training dry run measured $\mathcal{L}_{\text{CE}} = 2.434$ vs. $\gamma\mathcal{L}_{\text{SP}} = 99.12$ (ratio 40.7); the run proceeded

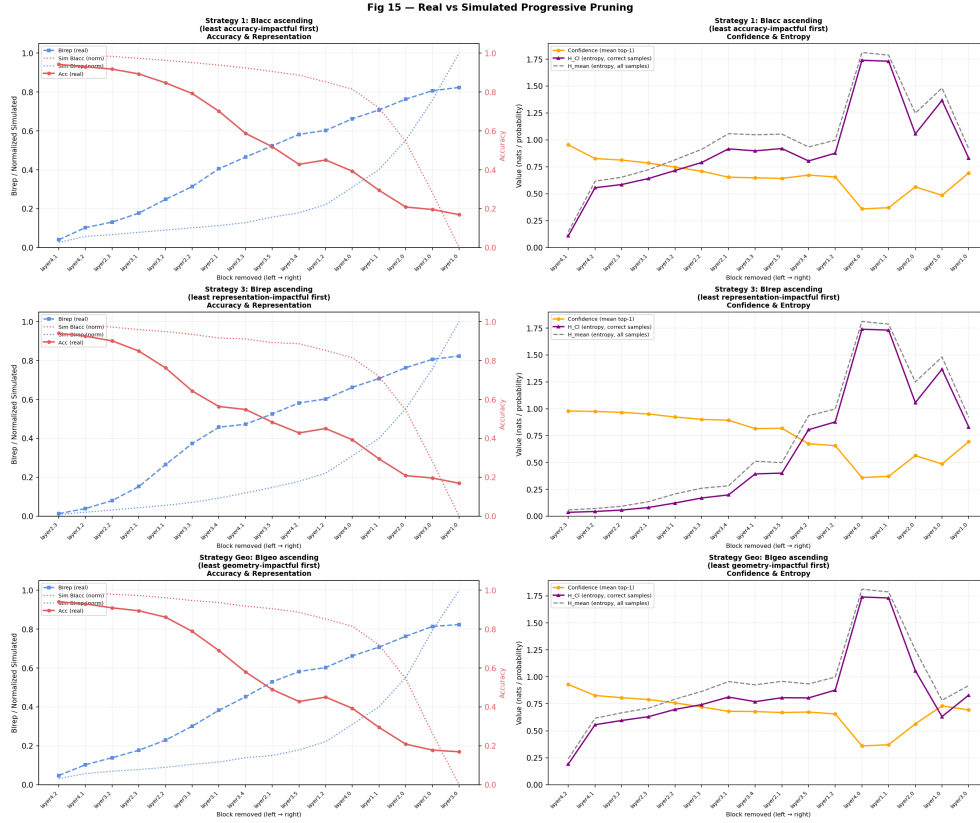


Fig. 5. Real progressive pruning, $k = 1, \dots, 16$ simultaneously ablated blocks, three removal orderings (rows). Left: real accuracy (red, right axis) and real BI_{rep} (blue) against the normalized additive predictions (dotted). Right: mean confidence and entropy traces. The widening gap between solid and dotted blue curves in the mid-range is the superadditivity finding; the H_{C1} spike at $k \approx 12-13$ is the population-level silent-failure regime.

because the warning threshold was set at ratio > 100 . During training, the SP term constitutes 88–92% of the total loss at epoch 1 and $\sim 99\%$ at epoch 200 in every SP-KD run. When one term is $\sim 99\%$ of the objective and the stage weights are constrained to sum to 1, changing how that term distributes over stages (between vectors that are 99.97% cosine-similar to begin with) cannot change what is being optimized. The weighting mechanism under test had effectively zero gradient leverage. An adaptive γ -halving guard was implemented after this diagnosis, but none of the twelve archived primary runs used it.

The Δ -weighted ablations (conf_gated, gated_only, delta_kd) were designed to decompose gating vs. soft-label KD on top of Δ -derived weights; as their artifacts were not archived (Section III-E), they support no quantitative claim. The single surviving record (conf_gated: accuracy 0.9511, final CKA 0.9811, loss split SP 85–88% / KD-soft 12–14% / CE $< 1\%$) comes from project session records rather than repository artifacts and is reported for completeness only. It motivates, without establishing, the inference that soft-label KD, rather than gating or stage weighting, is the primary driver of elevated teacher–student CKA.

E. Geometric fidelity confers no discriminative advantage

This is the substantive finding of Phase 4, and it runs against the project’s initial hypothesis. All numbers are seed-averaged over the archived artifacts.

Representational fidelity, as designed. SP-KD students track the teacher closely: per-class CKA vs. teacher 0.829–0.842, Frobenius norm of the class-cosine deviation 0.50–0.62 (vanilla: 0.468 and 1.156), and the smallest deviation of the confusion pattern from the teacher’s ($\|CM_{student} - CM_{teacher}\|_1$ off-diagonal: 116–121 vs. 163.7 for vanilla; Fig. 6).

The unfaithful student achieves the highest separation. Vanilla’s feature space is qualitatively different, not a degraded copy: intra-class variance 1.87 vs. teacher 5.39 and SP-KD 3.57–3.86 (Fig. 7); mean FDR over all 45 class pairs 15.70 vs. teacher 7.61 (ratio 2.06) and SP-KD 5.96–6.27 (ratios 0.78–0.82) (Fig. 8). On the critical pairs the picture is the same (cat–dog FDR: vanilla ≈ 10.3 , teacher 4.98, SP-KD ≈ 4.2 –4.3). The resolution of the apparent paradox (SP-KD keeps some centroids further apart yet scores lower FDR) is the denominator: vanilla collapses within-class dispersion, so its signal-to-noise ratio per pair is high even where centroids are close. Geometric fidelity to the teacher does not confer discriminative advantage; on this task it reduces it. The teacher’s diffuse geometry (ICV 5.39, PR 9.33) is plausibly an artifact of its 23.5M-parameter budget, not a property worth transferring to an 11.2M-parameter network, a specific instance of the fidelity-vs-student-quality gap of [10].

BI-Rep reproduces the teacher’s failure modes, with more confidence. On critical-pair confusions (counts summed over the six directional pairs, seed-averaged): vanilla 144.0, bi_acc

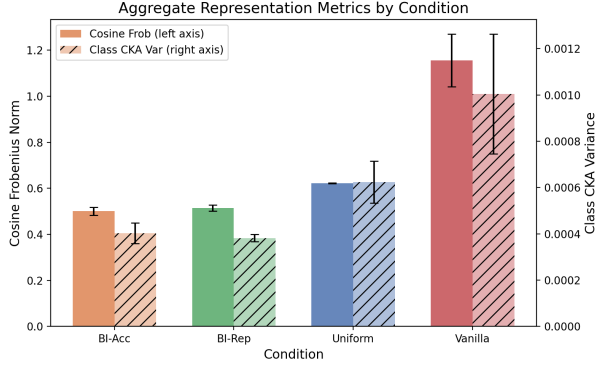


Fig. 6. Aggregate representational fidelity per condition: Frobenius norm of the class-cosine deviation from the teacher and per-class-CKA variance. BI-Acc and BI-Rep are lowest on both axes; vanilla is highest. This licenses one claim only: SP-KD achieves what it optimizes. It says nothing about whether fidelity is useful.

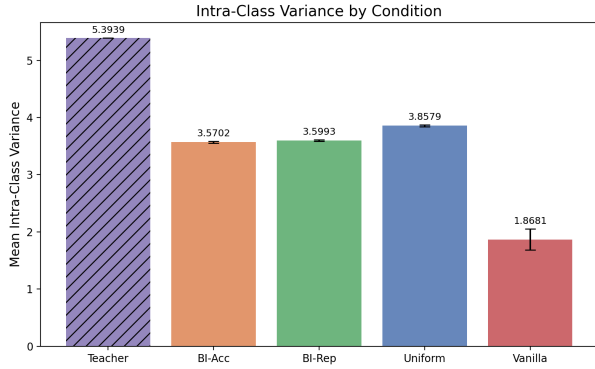


Fig. 7. Intra-class variance per condition against the teacher baseline. Vanilla (1.87) collapses clusters $\sim 3\times$ tighter than the teacher (5.39); SP-KD students (3.6–3.9) inherit the teacher’s dispersion, as their loss demands.

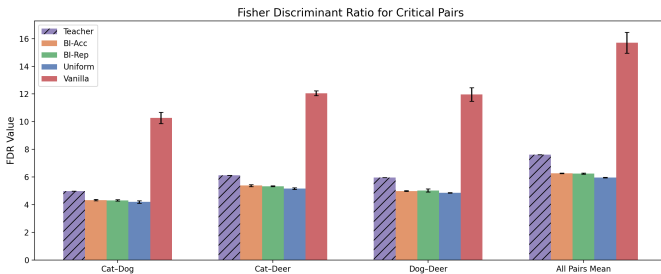


Fig. 8. Fisher Discriminant Ratio on the critical pairs (cat–dog, cat–deer, dog–deer) and the 45-pair mean; error bars over 3 seeds. Vanilla roughly doubles the teacher’s separation on every pair; all SP-KD conditions sit slightly below the teacher and on top of one another. Fidelity and separability dissociate.

145.3, bi_rep 151.3, uniform 152.3; the teacher itself commits 168. The condition designed to protect semantic structure (bi_rep) commits the most critical-pair errors among the BI conditions, and when it misclassifies dog as cat it does so at confidence 0.883, higher than bi_acc (0.870), vanilla (0.856), and the teacher itself (0.871). Students that copy the teacher’s geometry therefore inherit its confusions, with amplified confidence.

Participation ratio pins SP-KD students to the class count. Global PR: uniform 10.02, bi_acc 9.98, bi_rep 9.95, vs. vanilla

TABLE IV
ROBUSTNESS AND CALIBRATION. MCE: MEAN CORRUPTION ERROR NORMALIZED PER CORRUPTION BY THE TEACHER (TEACHER MCE $\equiv 1.0$, LOWER IS BETTER); RmCE: RELATIVE MCE; ECE ON CLEAN AND CORRUPTION-AVERAGED DATA.

Condition	mCE ($\downarrow$)	RmCE	ECE clean	ECE corrupted
vanilla	0.948 ± 0.012	0.959	0.0297	0.182
bi_rep	0.959 ± 0.010	0.989	0.0275	0.177
bi_acc	0.960 ± 0.004	0.990	0.0266	0.177
uniform	0.966 ± 0.007	0.996	0.0268	0.178

9.22 and teacher 9.33 (Fig. 9). One interpretation is that $N \times N$ Gram matching spreads variance uniformly across the ~ 10 class-discriminative directions that dominate the pre-classifier layer, driving effective dimensionality to the number of classes. This reading is consistent with all three values but was not independently tested; it is an interpretation, not a measurement.

Behavioral equivalence on clean data. Confidence on correct predictions 0.9869–0.9885, overall entropy 0.061 (vanilla) vs. 0.067–0.069 (SP-KD), logit margin 8.28–8.32 for all students (83–84% of the teacher’s 9.92), calibration gap 0.035–0.038 vs. teacher 0.043. No student exhibits a behavioral silent-failure signature relative to the others.

F. Robustness and calibration under distribution shift

Table IV and Fig. 10 summarize CIFAR-10-C (19 corruptions $\times$ 5 severities) and CIFAR-10-P. All students are more corruption-robust than the teacher, and vanilla, the least teacher-faithful and most cluster-collapsed student, is the most robust. This is the opposite of the hypothesis that inheriting the teacher’s geometry yields robustness; the differences among SP-KD conditions are within seed-level variability. One calibration nuance survives per-corruption inspection: bi_rep is the only condition whose mean ECE never exceeds the teacher’s on any of the 19 corruption types (bi_acc is worse on 1/19, uniform on 1/19, vanilla on 2/19), a consistency property rather than a magnitude advantage. On CIFAR-10-P, mean flip probability is 0.0329–0.0340 for all students vs. 0.0384 for the teacher (~ 0.001 inter-condition spread). All students are more prediction-stable than the teacher, and no condition separates.

V. DISCUSSION

A. What the null result does and does not license

It does license two conclusions. (i) On CIFAR-10 with a ResNet-18 student, stage-level BI weighting of SP-KD has no measurable effect relative to uniform weighting, a result explained after the fact by the weight-vector collapse and the loss-share measurements of Section IV-D. (ii) The initial hypothesis that teacher geometry is a normative target for a smaller student is false in this setting, on the direct evidence of Section IV-E (FDR, ICV, robustness, critical-pair confidence).

It does not license rejecting the BI framework itself. The block-level measurements of Sections IV-A–IV-C were never given a chance to act in Phase 4. Sixteen block scores were compressed into 4 stage means, a low-pass filter that specifically cancels the per-block Δ profile, whose largest component

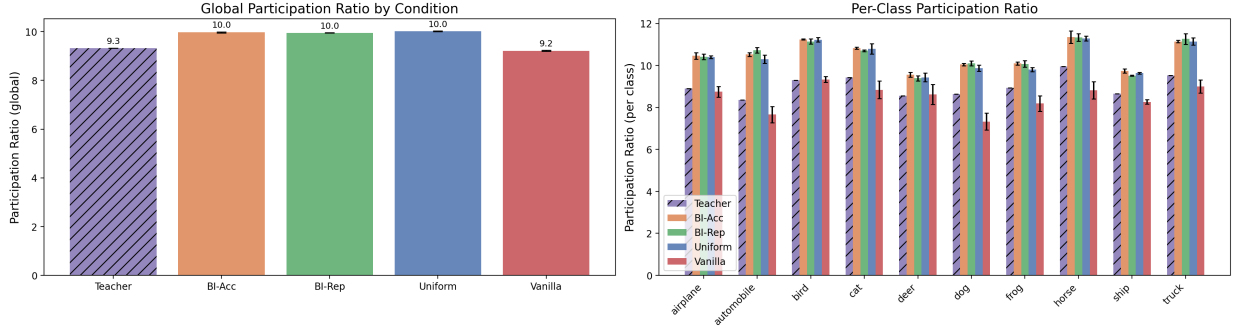


Fig. 9. Global and per-class participation ratio. All SP-KD conditions converge to $\approx 10 = n_{\text{classes}}$; vanilla and the teacher sit at ≈ 9.2 – 9.3 . The “uniform spreading over class directions” reading is an interpretation, not a measurement.

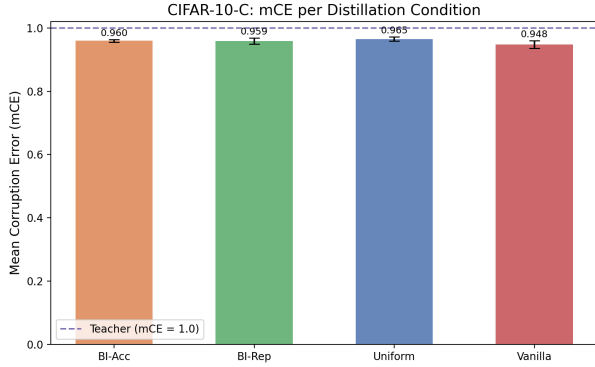


Fig. 10. Mean corruption error (teacher = 1.0). Vanilla is the most robust student; the SP-KD conditions cluster above it: geometry inheritance did not yield corruption robustness here.

(layer4.0) sits inside a stage otherwise made of near-zero- Δ blocks. The surviving 1.1-point weight differences were then scaled into a loss term whose share of the objective made all weightings equivalent. A setting with per-block matching, balanced losses, and task headroom (Section VI) could still test the prescriptive claim; symmetrically, nothing here validates the framework’s usefulness for distillation either.

The setting itself is saturated: vanilla reaches 0.950 and the best SP-KD seed 0.954. A weighting effect would have had at most ~ 0.003 of absolute accuracy to gain before hitting the observed ceiling, below the seed-noise floor (per-condition SD 0.0006–0.0024).

B. Limitations of the BI metrics

Marginality. All three metrics are single-block ablation scores. Section IV-C shows that joint damage exceeds the additive prediction by roughly a factor of two at $k = 9$; block importance interacts. This limitation is shared with ShortGPT and every marginal criterion in the literature, and the Δ criterion inherits it.

CKA caveats. Triangulation (Section III-C) reduces but does not remove the failure mode of [4]; and $N = 2500$ vs. $D = 2048$ is at the low end for Gram-spectrum stability (the design target was $N \approx 3000$ – 5000).

The limited utility of BI_{geo} . Its main virtue is cost (one forward pass); as a pruning guide it duplicates BI_{acc} ’s ordering

in practice (Section IV-C), and it gives no distinctive signal for the confirmed silent-failure block. A propagated variant that approximates ablation impact through the residual stream in a single pass was sketched but not implemented.

Known implementation blemish. The Phase-2 multilayer propagation profile selects its top-3 blocks by highest BI_{rep} rather than the intended highest Δ ; consequently layer4.0 has no propagation profile in the archived results. This affects one auxiliary analysis, not the primary metrics. The companion notebook omits this auxiliary profile entirely; no figure or table in this paper depends on it.

C. The methodological lesson

The $\gamma = 3000$ setting was calibrated once so that $\gamma \mathcal{L}_{\text{SP}} \approx \mathcal{L}_{\text{CE}}$ at initialization of an earlier configuration, and the dry-run guard warned only at ratio > 100 ; the measured ratio of 40.7 passed silently. The general lesson, stated as a requirement for any weighted-loss experiment, is to verify before training that the mechanism under test has non-negligible gradient leverage. Two checks suffice: (a) compare the candidate weight vectors themselves (here the two vectors were nearly identical, so the experiment was uninformative regardless of γ), and (b) track the loss-share, or better the gradient-norm share, of the weighted term over training against a hard budget. Both checks take minutes to run; the experiment they would have redirected consumed ~ 72 GPU-hours.

VI. FUTURE DIRECTIONS

A. Spectrally truncated SP-KD

The Phase-4 diagnosis suggests the problem is not that the student copies teacher geometry but *which part* of it. The teacher’s Gram matrix mixes class-discriminative structure with capacity-specific structure (the diffuse within-class dispersion a 23.5M-parameter model can afford). Proposal: match only the dominant spectral part of the teacher’s Gram matrix. Eigendecompose $G_T = \sum_i \lambda_i v_i v_i^\top$ and use the rank- k target $G_T^{(k)} = \sum_{i \leq k} \lambda_i v_i v_i^\top$ in Eq. (8), with k set by the student’s effective dimensionality (its participation ratio measured on held-out data; vanilla ResNet-18 lands at PR ≈ 9.2 here). Four conditions: (1) full-Gram SP-KD (the current uniform baseline); (2) truncated Gram, fixed k swept over $\{10, 50, 100, \dots\}$; (3) student-capacity-matched $k = \text{round}(\text{PR}_{\text{student}})$;

(4) a random-projection control onto a random rank- k subspace with matched spectral norm, testing whether any benefit comes from the specific top eigendirections or merely from reduced target complexity. Falsifiable predictions: condition $3 \geq 2 > 1$ on FDR and mCE, with condition 3 recovering part of vanilla’s separation advantage while retaining SP-KD’s calibration consistency; condition 4 at or below condition 1. A known risk is the rank-collapse feedback loop if k is estimated adaptively from the student’s current PR; mitigations are freezing k after a warm-up, a held-out PR estimate with a floor $k \geq n_{\text{classes}} + \text{margin}$, or annealing k on a fixed schedule. A novelty check against existing rank-constrained CKA/Gram-matching distillation methods is pending.

B. Non-saturated tasks and per-block matching

The other two design flaws have direct fixes. (i) *Task headroom*: on CIFAR-100 a ResNet-18 student does not saturate and the teacher–student gap is large enough for weighting effects to be measurable. (ii) *Granularity*: replace 4-stage matching with per-block matching, in which each of the 16 teacher blocks is paired to the nearest student block and carries its own BI weight (16 weighting points instead of 4, no stage averaging, no low-pass filter). This incurs no architectural cost because SP-KD Gram matrices are $N \times N$ and therefore dimension-agnostic: no projection heads are needed across arbitrary channel-width mismatches, which also opens cross-architecture teacher/student pairs (e.g., ResNet teacher $\rightarrow$ compact ViT student) not touched in this project.

VII. CONCLUSION

This project asked whether block importance should be measured by what its removal does to accuracy or to the representation, and whether the difference matters for distillation. The measurement phases show that the two views agree globally but dissociate locally, and that the dissociation can be behaviorally real: ablating layer4.0 leaves accuracy nearly intact while multiplying the entropy of correct predictions and merging class pairs. Real multi-block pruning shows that marginal scores bound joint damage from below by roughly a factor of two. The prescriptive phase produced a mechanistically diagnosed null result, in which the weighting mechanism was inert by construction, and an unexpected substantive finding: imitating a larger teacher’s geometry yielded no benefit and reduced separability and robustness in a half-capacity student. The methodological yield is a short pre-training checklist (weight-vector divergence, loss-share budget) that any weighted-loss experiment should pass before GPU time is spent on it.

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APPENDIX A

ARTIFACT AND REPRODUCTION MAP

Per the course guidelines, this appendix documents provenance and is not counted toward the word limit.

Every number in this paper is read from a repository artifact: `phase2_results.json` (all per-block metrics, baseline accuracy); `phase3_correlations.json`, `phase3_jaccard.json`, `phase3_silent_failure_layer4_0.json`, `phase3_per_class_cka.json`, `progressive_pruning_real.json` (Phase-3 analyses); `phase4_Vanilla/Uniform/BiAcc/BiRep_42/123/456.json` (final accuracy, CKA, loss curves, stage weights, dry-run statistics); `phase4_fine_analysis.json`, `phase4_extended_analysis.json`, `phase4_confidence_analysis.json` (geometry and behavior); `phase4_cifar10c_results.json`, `phase4_cifar10p_results.json` (robustness).
Reproduction:

```
pip install -r Bi_project/requirements.txt
cd Bi_project/code
python run_pipeline.py --phase all --device cuda
python run_pipeline.py --phase4 \
    --phase4-conditions \
        vanilla uniform bi_acc bi_rep \
    --phase4-seeds 42 123 456
python phase4_analysis.py
python phase4_analysis_extended.py
python phase4_confidence_analysis.py
python phase4_cifar10c_eval.py --dataset both
```

Alternatively, the companion Colab notebook reproduces all figures and tables end-to-end on a T4 GPU (~60 min excluding Phase-4 retraining; Phase-4 figures are produced from the embedded snapshot of the archived artifacts, with full from-scratch training code included behind a flag). Documented provenance caveats: the `conf_gated` condition’s outcome survives only in project session records (no JSON/checkpoint); the `gated_only` and `delta_kd` conditions are implemented but have no outcome record; no test suite survives in the repository (validation relies on the runtime checks described in Section III).

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